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Determination of the property evolution in rock after laser irradiation by nanoindentation and nuclear magnetic resonance

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ABSTRACT

Laser technology, as a novel approach to rock breaking, holds great promise for accessing resources in deep, hard rock layers. Therefore, it is crucial to conduct microscopic testing and analysis of rocks following laser irradiation. In this study, the changes in minerals and elements of the rocks after laser irradiation are clarified using X-ray diffraction (XRD) and energy dispersive spectroscopy (EDS). Additionally, the primary temperature range for rock weight loss is determined through thermogravimetric analysis (TGA). The rocks' microscopic features after irradiation are observed using a scanning electron microscope (SEM) and nuclear magnetic resonance (NMR), while the micromechanical properties of the rocks after irradiation are analyzed using nanoindentation techniques. New components such as $Ca(OH)_2$ and CaO are generated in the laser irradiation zone. There is a decrease in the mass percentage of carbon (C), from 10.10 % without laser irradiation to 9.70 % with irradiation, and a decrease in the mass percentage of oxygen (O), from 46.38 % without irradiation to 43.99 % with irradiation. The percentage of large and medium-sized pores is evenly distributed, accounting for more than 50 % of the total after laser irradiation. The small pores occupy 18.85 % and 18.08 % within the size range of 0.0025–0.01 μm and 0.01–0.025 μm, respectively, accounting for a combined total percentage of slightly over 35 %. In the nanoindentation test, the laser irradiation area exhibits more significant deformation, with an average maximum indentation depth increase of 139.4 % compared to non-irradiated areas. The average microscopic modulus of elasticity of the rocks decreases from 36.69 GPa without laser irradiation to 10.06 GPa with laser irradiation, representing a reduction of 72.58 %. Moreover, the average hardness of the rocks decreases from 1.67 GPa without laser irradiation to 0.47 GPa with laser irradiation, indicating a decrease of 71.86 %. This study provides valuable foundational data for potential engineering applications.

1. Introduction

The primary battleground for oil and gas exploration and development has progressively shifted towards unconventional formations, which are intricate and challenging to drill.[1–3] This shift has made drilling considerably more demanding. Enhancing drilling speed through the utilization of new materials and structural improvements of the drill bit alone is insufficient when dealing with intricate and challenging formations, particularly ultra-deep and highly abrasive ones.[4–6] Hence, it is imperative to develop highly efficient technologies for rock breaking. Emerging drilling methods, including laser,[7,8] electric pulse,[9,10] microwave,[11,12] plasma,[13] and thermal laminar fracturing,[14] are continuously evolving to address the challenges posed by hard rock formations. Among these methods, laser rock breaking is widely regarded as the most effective.

In 1917, Albert Einstein made a prediction about the potential occurrence of excited radiation, which involves the generation of photons or discrete beams of energy through transitions between atomic or molecular energy levels.[15] In 1960, Maiman made the world's first excitation (ruby) light machine based on Einstein's theory.[16] In 1968, the Massachusetts Institute of Technology and the University of California jointly proposed the concept of utilizing laser drilling for rock breaking.[17] In 2012, FORO integrated laser equipment with conventional mechanical drilling tools and conducted a series of laboratory

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experiments on laser-mechanical rock breaking.[18] Since the inception of the idea to employ lasers for rock breaking, laser rock breaking technology, including both laser direct and laser-assisted rock breaking, has been in a continuous state of development, and this progress continues to this day.[19–21]

Laser rock breaking is an intense process that employs lasers to produce highly concentrated, high-energy laser beams directed at the rock surface. This concentrated energy results in the rock weakening, fracturing, melting, and vaporizing due to the localized high temperatures, often accompanied by thermal bursting phenomena. Gahan and Samih et al.[22] experimented with sandstone, limestone, and shale using a 1.6 kW Nd: YAG laser. The results indicated that the specific energy increased as the irradiation time increased, with shale samples exhibiting the lowest specific energy values when compared to limestone and sandstone samples. Graves et al.[23] conducted experiments with four lasers, CO_2, CO, COIL, and Nd: YAG, with powers ranging from 0.5 kW to 10 kW. Lasers using CO_2, CO, and Nd: YAG in pulsed mode were found to be more effective in rock destruction compared to COIL lasers with continuous transmission. Xu et al.[24,25] used a pulsed Nd: YAG laser to irradiate the rocks. The test data indicated that the permeability of the rocks subjected to pulsed Nd: YAG laser irradiation increased by 566 % in comparison to the untreated rocks. The effect of pressure on the thermal action of laser rocks was analyzed by Gahan and Batarseh.[26] It was observed that the removal mechanism of sandstone involves exfoliation, where the rapid disparity in thermal expansion leaded to the fracturing of particles and cementitious material. Mahdi Bakhtbidar et al.[27] conducted a study on the penetration of laser-irradiated rocks using a CO_2 laser beam. It found that the CO_2 laser was effective in penetrating the rocks when compared to other types of high-powered lasers. Ahmadi et al.[28] investigated the effect of the surrounding pressure on the destruction of rocks by laser irradiation. As the surrounding pressure increased from 8 to 16 MPa, the specific energy increased rapidly while the drilling rate decreased rapidly. Hu et al.[29] conducted experiments on laser irradiation of rocks using a 6 kW fiber laser. It was discovered that the size of the rock had minimal impact on its specific energy. Li et al.[30] used a high-power density laser for rock crushing of sandstone with different laser parameters. A complex network of cracks developed on the surface of the sandstone specimens, and both the crack area and length exhibited nearly linear increases with the number of laser perforations. Jamali et al.[31] demonstrated the application of high-power lasers in rock crushing and drilling processes by laser-induced rock softening. Shi et al.[32] proposed a new method of rock breaking by combining pulsed laser and induced cavitation, which has the potential for low energy consumption and high efficiency in rock breaking. The study examined the characteristics of circumferential strain, temperature, pore size, and modification specific energy of shale under laser irradiation with varying power, frequency, and focal length.[33] Pan et al.[34] investigated the variation in rock breaking efficiency with different numbers of holes and laser power while maintaining a constant total laser irradiation energy.

In order to elucidate the fundamental principles of rock breaking through laser technology, scholars have conducted comprehensive investigations of rock specimens employing advanced testing methodologies. Masaaki Sakakura et al.[35] investigated the structural changes inside rock salt crystals after femtosecond laser irradiation using microscopic pump-probe techniques and elastic simulations. Yan et al.[36] conducted laser irradiation experiments on sandstone with laser power ranging from 1 to 4 kW. It observed that the vapor/plasma generated during laser perforation underwent an abrupt change in the process, as evidenced by high-speed video images. Additionally, X-ray diffractometer analysis indicated that the rock phase after perforation was primarily composed of SiO_2 and Fe_2O_3. Li et al.[37] conducted an experimental study for the fracture mechanism of laser-irradiated granite. SEM analysis confirmed that the fracture mechanism of granite under laser irradiation was a typical brittle fracture. Experiments on laser irradiation of limestone at different powers were carried out by Wang et al.[38] Three-dimensional X-ray microimaging (XRM) images revealed that the rock shot holes exhibited a V-shaped structure with cracks extending throughout the entire specimen. Homand-Etienne et al.[39] studied the phenomenon of thermal cracking in granite samples in the temperature range of 20–600 °C. Observations made using scanning electron microscopy (SEM) revealed that the length of cracks in granite remained relatively unchanged, but their width increased as the cracking temperature increased. Batarseh, Gahan et al.[40,41] conducted experiments on continuous laser irradiation of rocks, combining microscopic techniques such as CT, XRD, and SEM to analyze the changes in rock microstructure (e.g., fracture extension and melting) under laser irradiation. Bazargan et al.[42] conducted a study on the formation of fine and large fractures by profiling the rocks before and after laser irradiation operations. They employed various analyzers such as spectrophotometers and advanced electron microscopy for their research.

With the ongoing increase in the depth of energy extraction, traditional drilling methods have become inadequate to meet the growing demand. Laser technology, as a promising approach to rock breaking, has been the subject of extensive research by numerous scholars. However, prior experimental studies have primarily emphasized the macroscopic extension of fractures and changes in rock breaking efficiency. Few comprehensive microscopic analyses of rocks following laser irradiation have been conducted to date. Therefore, this study will employ XRD, EDS, TGA, SEM, NMR, and nanoindentation techniques to scrutinize the effects of laser irradiation on rocks, encompassing aspects such as mineral composition, elemental content, weight loss temperature, microscopic morphology, pore count, and micromechanical properties. Moreover, this paper represents the inaugural experimental investigation in which nanoindentation technology has been employed on rock by laser irradiation. The data acquired and the detection outcomes hold immense significance for both the practical application and physical modeling of laser-induced rock fragmentation.

2. Experiment procedure

2.1. Samples preparation

The samples used in this study are excavated from an outcrop of limestone in Sichuan Province, China. These samples are then shaped into standard specimens with dimensions of Φ50 mm × 50 mm. The rock should be heated at 155 °C for 24 h to eliminate moisture before starting the laser irradiation experiment. Limestone is a carbonate rock with calcite as its main component. It is one of the most widely distributed rocks in the earth's crust. The physical and mechanical properties of limestone are shown in Table 1.

2.2. Apparatus and analysis method

Fig. 1 shows the laser irradiation system, which consists of a laser generator, a digital laser console, a focusing system, an air compressor, a storage cylinder, an infrared thermal imager and a computer. The laser generator model QCW-150W provides a maximum of 150 W laser power. An air compressor and storage cylinder provide auxiliary gas to prevent slag damage to the laser lens as well as to aid ignition. During the experiments, the switching on and off of the laser generator and auxiliary gas and the position of the laser head are controlled via a

Table 1
Physical and mechanical properties of limestone.

Density, g/cm^3	Poisson's ratio	Uniaxial compressive strength, MPa	Tensile strength, MPa	Porosity (%)	Heat expansion coefficient, m/m K
2.56	0.26	68.45	8.59	4.1	7.9×10^{-6}

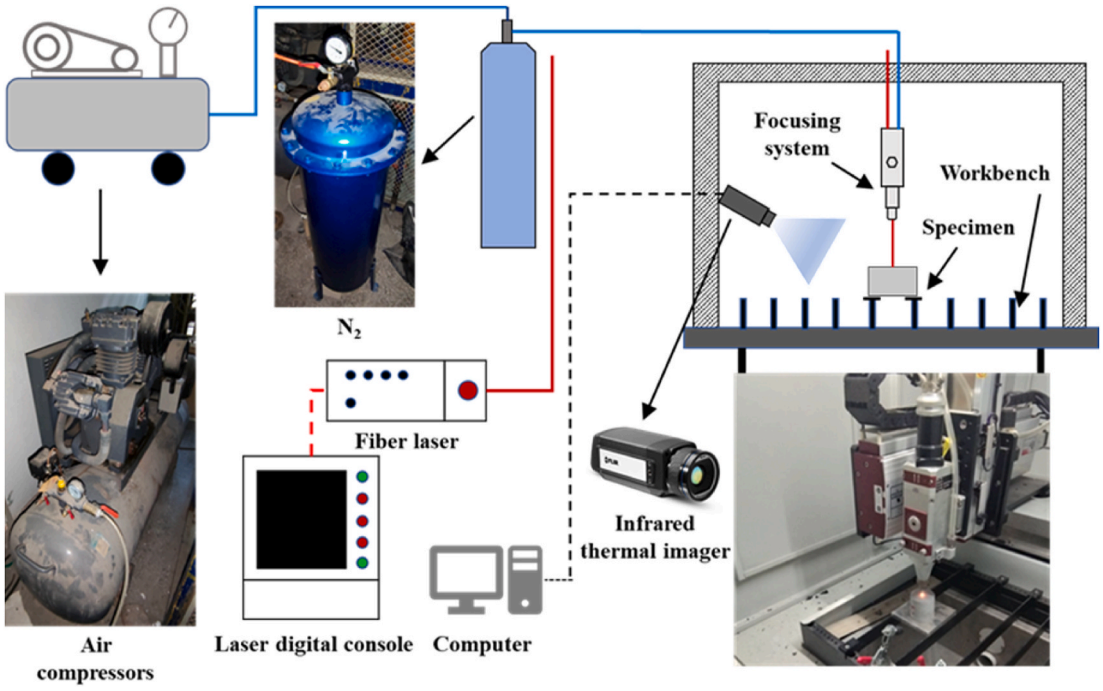


Fig. 1. Laser irradiation systems.

digital laser console. The temperature of the specimen surface is recorded during laser irradiation using an infrared thermal imager. The experimental parameters used for rock irradiation by laser are detailed in Table 2.

The analysis method for rocks after laser irradiation, as depicted in Fig. 2, involves three main aspects.

(1) Mineral Composition and Elemental Analysis: This aspect focuses on determining the mineral composition of the rocks and analyzing their elemental content.
(2) Microscopic Morphology and Pore Size Distribution: Here, the microscopic features and distribution of pores within the rocks are examined to understand their morphology and pore sizes.
(3) Micromechanical Properties: This perspective involves the analysis of the micromechanical properties of the rocks, such as their hardness and elasticity, at a small scale.

These analyses provide a comprehensive understanding of how rocks are affected by laser irradiation, encompassing their chemical composition, physical characteristics, and mechanical behavior.

2.3. Nanoindentation testing

The nanoindentation instrument is a device used to measure the mechanical properties of the sample employing a diamond indenter with well-defined geometrical and mechanical characteristics, as depicted in Fig. 3. The equipment is primarily utilized to measure hardness and elastic modulus at the nanometer scale. It can also be employed for the study or testing of various properties, including contact stiffness, creep, elastic work, plastic work, fracture toughness, stress-strain curves, fatigue, memory modulus, and loss modulus of nanomaterials such as thin films. Additionally, it is suitable for the detection and analysis of a wide range of materials, whether organic or inorganic, soft or hard. The matrix materials can encompass a broad spectrum, including metals, alloys, semiconductors, glass, minerals, and organic materials.

As depicted in Fig. 4(a), the nanoindentation technique involves indenting a diamond indenter with known geometry (such as trigonal, spherical, etc.) and mechanical parameters (including elastic modulus and hardness) into a material with unknown mechanical properties. This process yields a load-displacement (F-h) curve for the loading-unloading sequence, as shown in Fig. 4(b). The elastic modulus (E) and hardness (H) of the specimen can be determined through subsequent analysis. During the indentation process, as the indenter is pressed into the surface of the sample, the load (F) is continuously increased until it reaches the peak load of the test. Subsequently, the load is unloaded until it reaches 0. The initial unloading slope of the applied loading force (F) and the indentation depth (h) is defined as the contact stiffness (S)[43]:

$$S = \frac{\mathrm{dF}}{\mathrm{d}h}\bigg|_{h_{\max}} \tag{1}$$

where h_{max} is the maximum pressing depth. F is the loading force. The converted modulus of E_r is described as[44]:

$$E_r = \frac{\sqrt{\pi}}{2\gamma\sqrt{A_c}} S \tag{2}$$

where, γ is the dimensionless correction factor related to the shape of the indenter, and A_c is the projected contact area of the indenter on the sample surface, which is usually a function of the maximum contact depth h_c, $h_c = h_{max} - \lambda P/S$ (λ is a constant, depending on the tip of the indenter).

The relationship between the elastic modulus of the sample and the converted modulus of the indenter can be expressed as[45]:

Table 2
Laser irradiation parameters.

Laser Model	Output power	Focal length	Irradiation distance	Irradiation time	Laser frequency	Whether to blow slag
QCW-150W	135 W	3.8 mm	4.0 mm	110 s	200 Hz	Yes

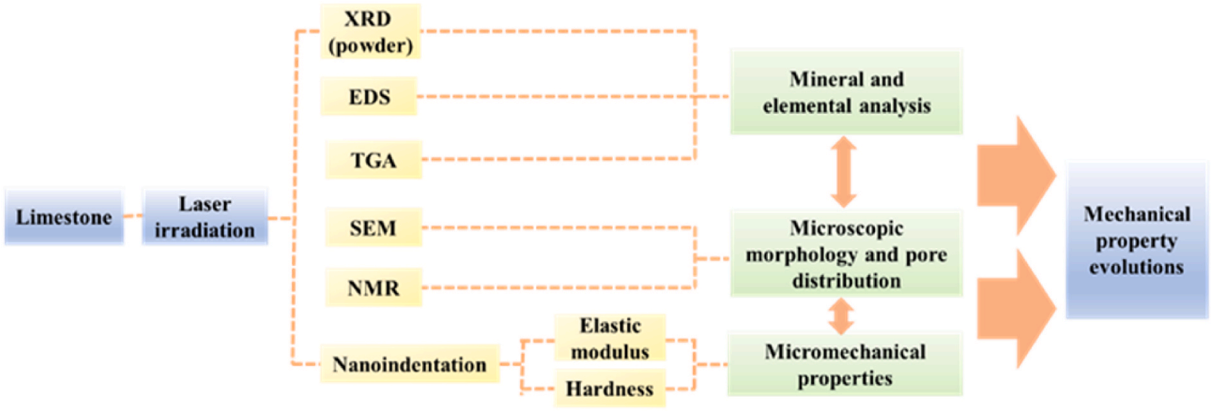


Fig. 2. Schematic diagram of the experimental methodology.

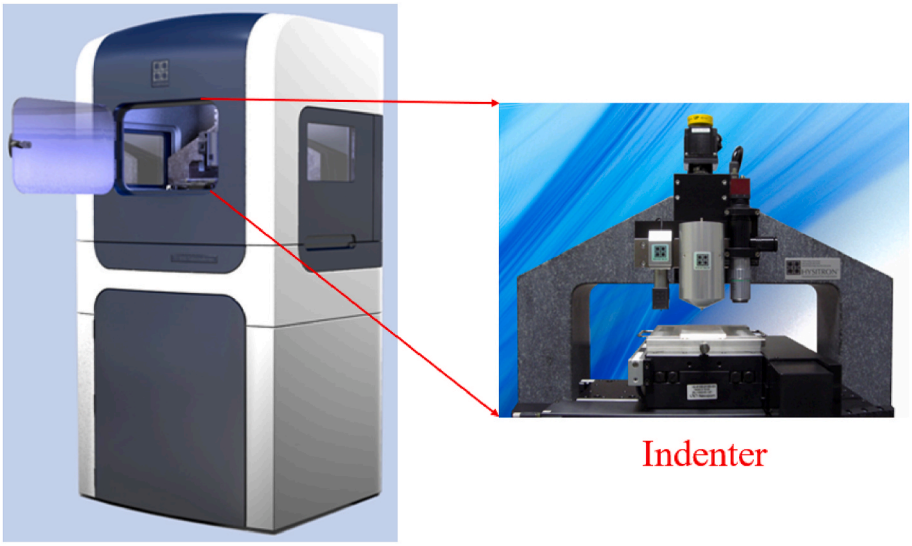


Fig. 3. Diagram of nanoindentation instrument.

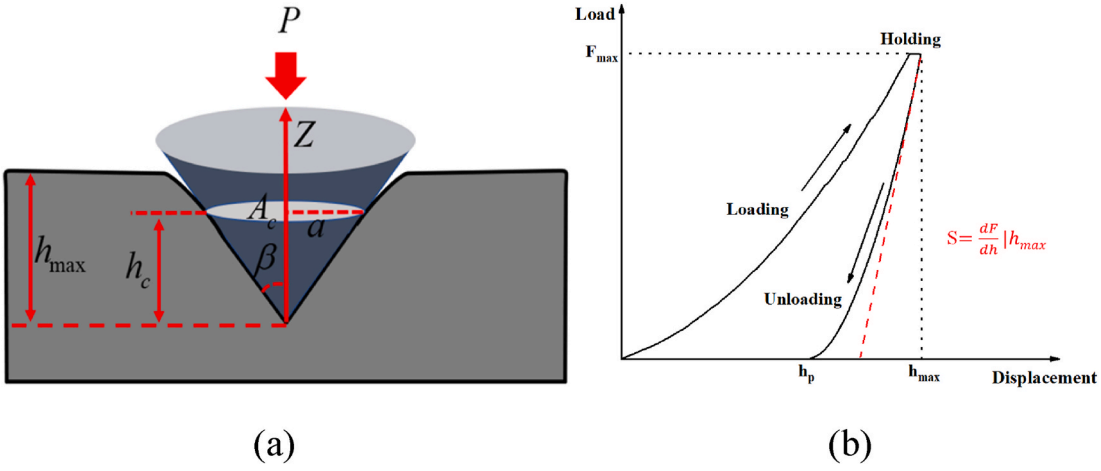


Fig. 4. (a) Schematic diagram of nanoindentation; (b) Typical load-displacement curve.

$$E = (1-\nu^2)\left[\frac{1}{E_r} - \frac{1-\nu_i{}^2}{E_i}\right] \quad (3)$$

where ν and ν_i are Poisson's ratio of sample and indenter, respectively. E and E_i are the elastic modulus of the sample and indenter respectively.

2.4. Thermogravimetric analysis

Thermogravimetric analysis (TGA) is a technique used to measure the change in the mass of a substance as a function of temperature or time under programmed temperature conditions. Its primary feature is

its robust quantitative ability, enabling accurate measurement of material mass changes and their rates, facilitating the detection and analysis of both physical and chemical processes. This equipment can be applied across a wide range of fields including polymer materials science, inorganic materials, mineral mixtures, metal materials, ceramics, food, chemical industry, and pharmaceuticals. Its capabilities encompass characterizing the thermal properties of materials and providing quantitative information on the composition and thermal stability of a variety of materials. Here are some specific details about its capabilities.

(1) Temperature range: room temperature - 1100 °C;
(2) Test gases: nitrogen or air;
(3) Heating rate: 0.1–250 °C/min;
(4) Equipment sensitivity: 0.1 μg.

2.5. Nuclear magnetic resonance

The nuclear magnetic resonance (NMR) imaging analyzer has the capability to perform geotechnical analysis and imaging functions. Additionally, it can be outfitted with various self-developed hardware modules, such as low-temperature and high-pressure control modules, as well as high-temperature and high-pressure function modules. These hardware additions enable the system to conduct simulation studies under a wide range of temperature and pressure conditions. This versatile setup allows for comprehensive research and analysis in geotechnical and related fields. Furthermore, it finds extensive applications in various fields including geotechnical engineering, geological research, energy exploration, and more. The technical parameters of the equipment are as follows: 2 MHz/21 MHz optional; Type of magnet: permanent magnet; Magnetic field strength: 0.3 ± 0.03 T/0.5 ± 0.03 T; Main frequency: 12 MHz/21 MHz; Probe coil diameter: 60 mm.

3. Results

3.1. Macroscopic failure and temperature distribution

Fig. 5 illustrates the change in the rock surface before and after laser irradiation. As depicted in the figure, white material is generated near the laser irradiation hole, and there is also some rock spalling near the small hole. Additionally, the rock surface exhibits the formation and extension of three cracks originating from the vicinity of the laser irradiation holes and extending towards the edge of the rock.

Fig. 6 displays the temperature distribution of the specimen during laser irradiation for durations ranging from 10 s to 110 s. Observing the figure, it becomes evident that the highest temperature on the specimen's surface is concentrated at the center of the irradiation. The highest temperature is observed at 10 s, with a recorded temperature reaching nearly 1600 °C. This phenomenon is attributed to the Gaussian distribution of the laser beam, where the highest energy density is concentrated at the center, gradually decreasing towards the periphery. Hence, it follows that the temperature of the object increases with higher energy density. The temperature peaks at the center of illumination at 10 s and subsequently diminishes as time progresses. This phenomenon is attributed to material vaporization at the center, resulting in the formation of small holes. As the laser penetrates deeper, the temperature at the center of the specimen's surface decreases. Additionally, it's worth noting that the area on the specimen's surface where the temperature exceeds 100 °C gradually expands over time. Indeed, this phenomenon can be attributed to the continuous transfer of heat as the specimen undergoes laser irradiation. Heat is continually transferred outward from the laser's point of contact, causing the region with a sample surface temperature exceeding 100 °C to expand.

3.2. Variation in mineral composition

The mineralogical composition changes in irradiated rocks are determined through XRD analysis. Fig. 7(a) and (b) display XRD spectra of samples in the non-irradiated area and the laser irradiation area, respectively. The primary mineral in the original rock sample is calcite, evidenced by a diffraction peak at approximately 3250, as depicted in Fig. 7(a). Additionally, small amounts of quartz and dolomite are detected in the original rock sample. Fig. 7(b) illustrates the diffraction intensity of rock samples irradiated by laser. It is evident that the diffraction intensity of the laser-irradiated rock is significantly lower than that of the original rock. The primary mineral in the sample remains calcite, with a diffraction peak at around 1300. As the temperature in this region exceeds 900 °C, calcite undergoes a transformation into CaO and CO_2, resulting in the presence of lime in the irradiated area. Additionally, portlandite is found in the laser-irradiated area. This can be attributed to the fact that the lime in the powdered sample used for the XRD test absorbs moisture from the air and becomes $Ca(OH)_2$.

Table 3 presents the alterations in the mineral composition of the rock between the original rock and the laser-irradiated area. It can be observed that the primary rock minerals predominantly consist of quartz, calcite, dolomite, and other amorphous or very low-content materials, with mineral proportions of 2.1 %, 92.2 %, 0.7 %, and 5.0 %, respectively. The mineral composition of the laser irradiation area includes quartz, calcite, portlandite, calcium oxide, and other materials, with respective proportions of 1.6 %, 26.2 %, 10.4 %, 1.8 %, and 60.0 %. The laser irradiation has resulted in a reduction of quartz content by 0.3 % and a significant decrease of calcite content by 66.0 %. Additionally, the dolomite has undergone thermal decomposition. New components such as $Ca(OH)_2$ and CaO have been produced in the irradiated area. This difference in behavior is due to the fact that both dolomite and

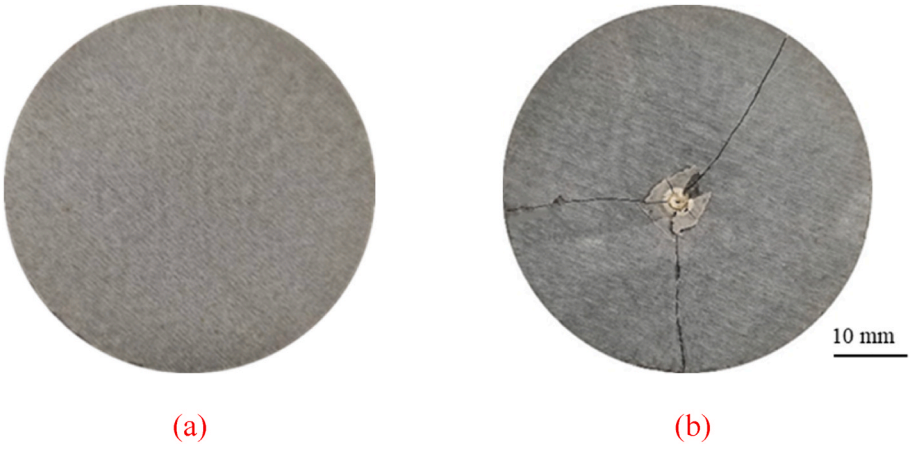


Fig. 5. Surface morphology of the specimen. (a) Before laser irradiation; (b) After laser irradiation.

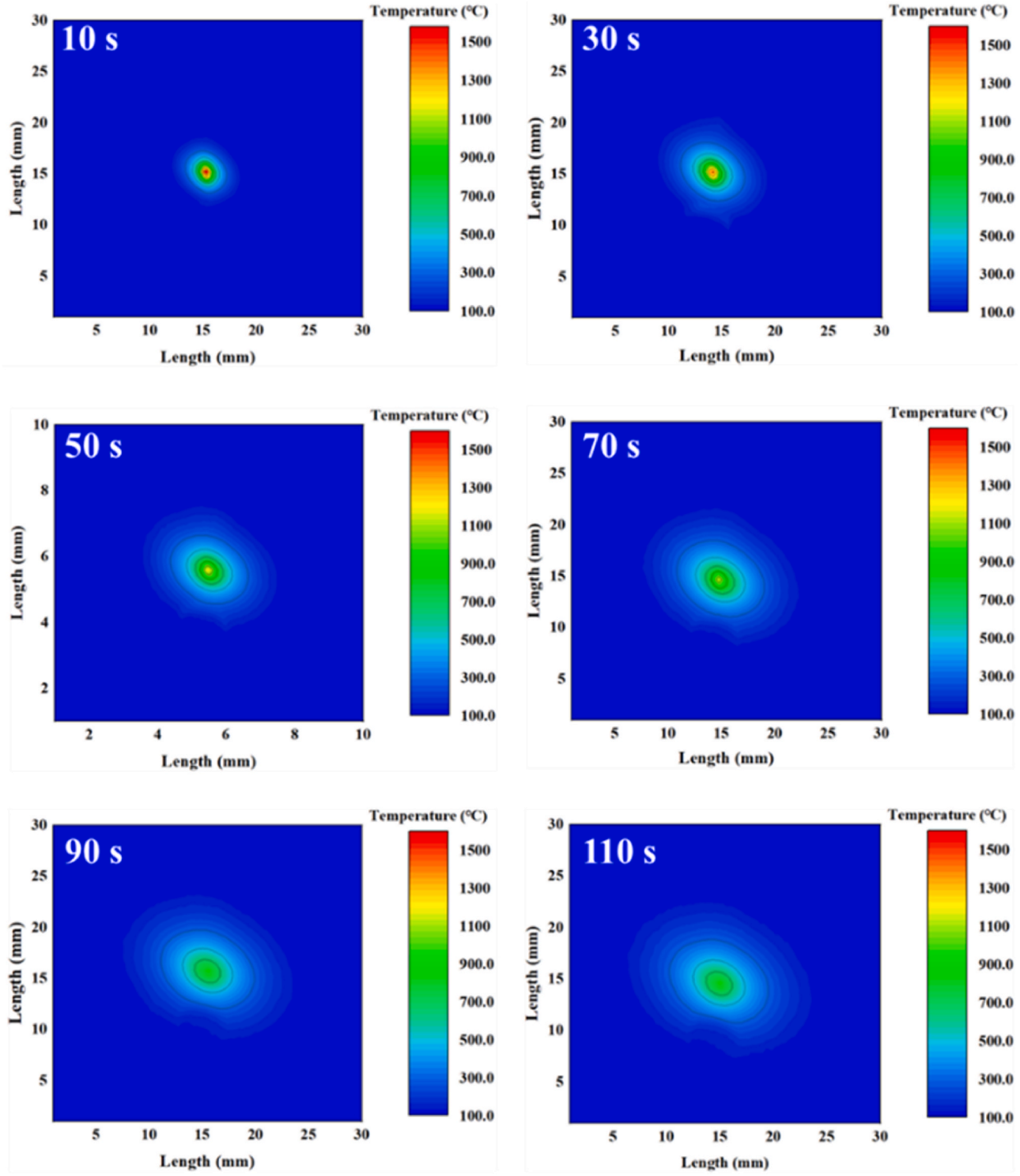


Fig. 6. Surface temperature distribution of specimens at different times.

calcite are carbonate minerals, but dolomite begins to decompose at around 650 °C, while calcite starts to decompose into CaO and CO_2 at temperatures of around 850 °C. After laser irradiation, the mineral composition lacks dolomite, and the calcite content is significantly reduced. Quartz, on the other hand, does not undergo significant changes in content since it requires temperatures above 1750 °C for thermal decomposition, while the center of laser irradiation typically reaches temperatures of around 1600 °C. Moreover, the calcium oxide produced by thermal decomposition can react with moisture in the air to form $Ca(OH)_2$.

3.3. Variation of elemental composition

During laser irradiation of rocks, an interaction occurs with the rock minerals. Specimens located near the irradiated holes are selected and tested for mineral elements to quantify the damage caused to the rock by laser irradiation. Fig. 8(a) and (b) display the energy dispersive spectrometer full spectrum analysis of the samples in the laser non-

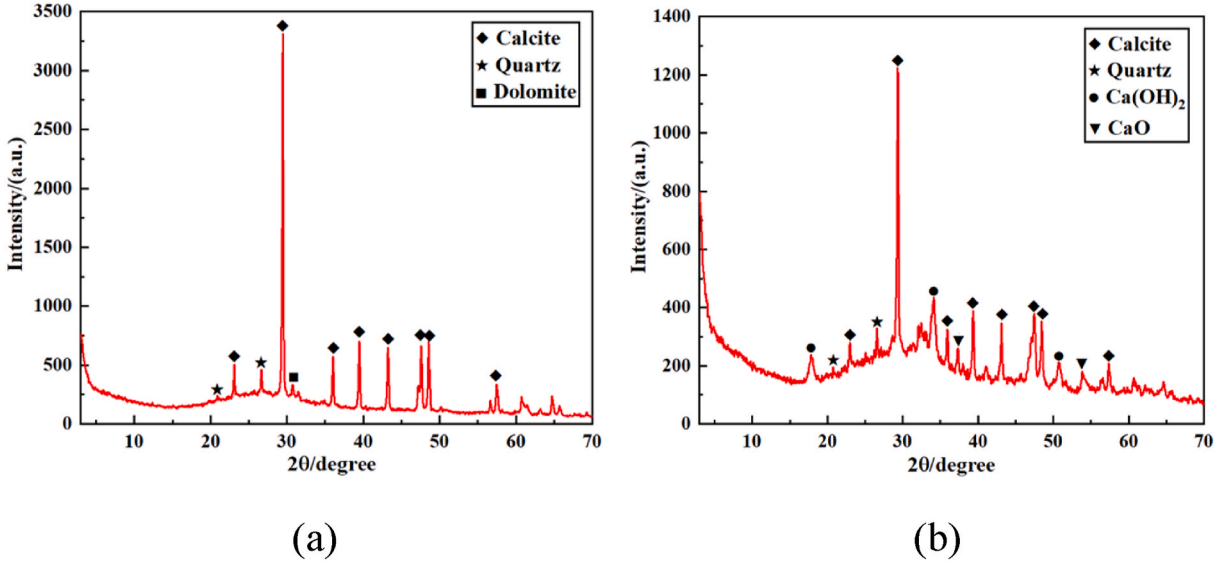


Fig. 7. XRD spectra (a) laser non-irradiation area; (b) laser irradiation area.

Table 3
Change of mineral composition.

Zone	Quartz	Calcite	Dolomite	Portlandite	CaO	Others
Original rock zone	2.1 %	92.2 %	0.7 %	–	–	5.0 %
Laser irradiation zone	1.6 %	26.2 %	–	10.4 %	1.8 %	60.0 %

irradiation area and laser irradiation area, respectively. It can be observed from Fig. 8(a) that the rock sample near the laser irradiation zone is enriched with Ca elements. The enriched area also contains major elements such as C and O, alongside small amounts of minor elements such as Mg and Si. From Fig. 8(b), it can be observed that the rock sample before laser irradiation is enriched with Ca near the irradiated area, which also contains major elements such as C and O, along with small amounts of minor elements like Mg and Si. However, the peak values of each element are reduced compared to the before laser irradiation levels.

Specimens from both the laser non-irradiation area and the laser irradiation area are chosen for quantitative mineral element testing. Table 4 clearly indicates a notable reduction in the relative concentrations of both C and O in the samples from the inner walls of the irradiated holes. The initial concentrations of C and O, which are 2.13 and 10.39, respectively, before irradiation, have decreased to 1.76 and 8.99. This phenomenon occurs because the laser irradiates the rock, causing the calcium carbonate compounds within the rock to decompose at high temperatures, producing CO_2, which escapes into the surrounding air environment. This results in a reduction in the apparent concentration of C and O. There is also a slight reduction in Mg and Si in the sample after laser irradiation of the rock, indicating that high-temperature irradiation of the rock can indeed have some impact on the minerals near the irradiated holes. As can be seen from the mass percentage row in Table 4, there is a significant decrease in both C and O. The mass percentage of C decreases from 10.10 % before irradiation to 9.70 % after irradiation and the mass percentage of O decreases from 46.38 % before irradiation to 43.99 % after irradiation. The mass percentages of Mg and Si have also decreased, with Mg decreasing from 0.45 % to 0.39 % and Si decreasing from 2.59 % to 1.48 % after irradiation, respectively. The

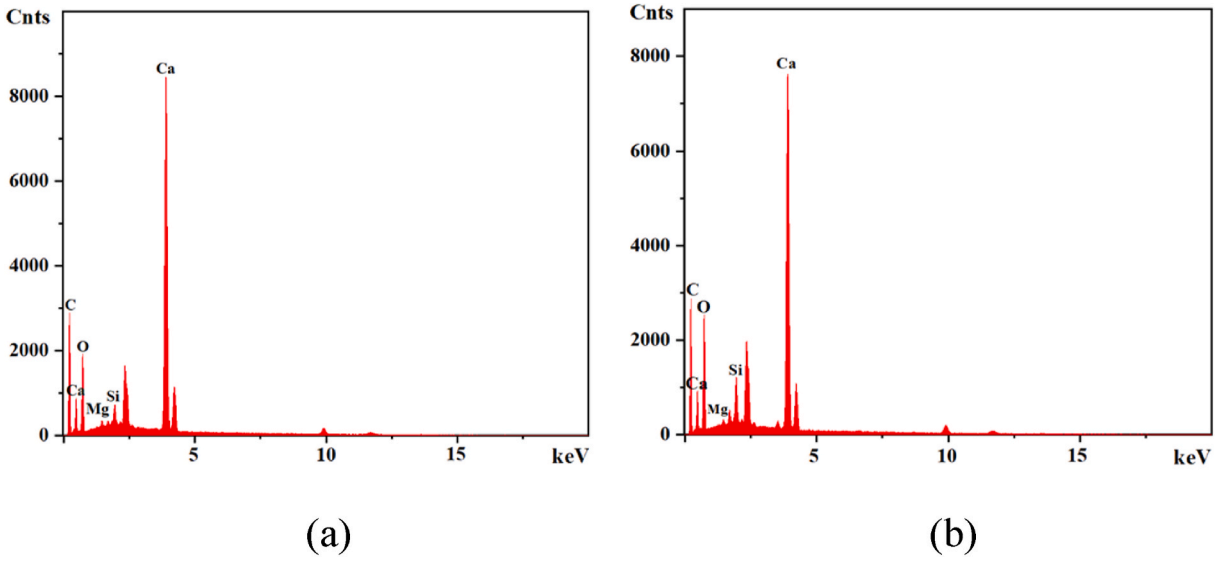


Fig. 8. EDS spectra (a) laser non-irradiation area; (b) laser irradiation area.

Table 4
Comparison of mineral element changes in the original rock zone and the laser irradiation zone.

Projects		C	O	Mg	Si	Ca
Laser non-irradiation area	Apparent concentration	2.13	10.39	0.18	1.33	24.27
	wt%	10.10	46.38	0.45	2.59	40.48
	Percentage of atoms	17.26	59.49	0.38	1.89	20.98
Laser irradiation area	Apparent concentration	1.76	8.99	0.15	1.18	26.52
	wt%	9.70	43.99	0.39	1.48	44.44
	Percentage of atoms	17.03	58.01	0.33	1.11	23.51

high-temperature region may lead to a reduction in Mg and Si elements, while the mass percentage of Ca increases from 40.48 % before irradiation to 44.44 % after irradiation. This is because Ca is mainly present in the sample as the compound CaO, whose boiling point needs to reach 2850 °C. Laser irradiation cannot reduce its evaporation, but other elements are reduced, resulting in an increase in the mass percentage of Ca.

3.4. Variation of mass with temperature

Fig. 9 shows the thermogram curves and the weight loss rate as a function of temperature obtained from the experiment. It can be seen that the weight loss of the samples first decreases, reaching an inflection point at 859 °C, and then gradually increases to reach the equilibrium position. The whole experiment is performed in a nitrogenous environment and the cause of the weight loss is volatilization of small molecules and decomposition of some minerals. The main weight loss temperatures of the samples occur in the range of 640–900 °C. This is because dolomite and calcite are thermally broken down in this temperature range, and their thermal decomposition reaction is described as follows:

$$CaMg(CO_3)_2 \rightarrow CaO + MgO + 2CO_2 \text{ (Dolomite)}$$

$$CaCO_3 \rightarrow CaO + CO_2 \text{ (Calcite)}$$

3.5. Observation of microscopic morphology

Scanning electron microscopy is employed to observe precise microscopic morphological features of the rock samples. The corresponding rock samples are made into small pieces of less than 1 cm × 1 cm × 1 cm, and the conductive layer is sprayed on the surface of the rock samples for direct analysis and characterization. The original rock sample and the irradiated region are observed sequentially. SEM images of different parts are shown in Fig. 10.

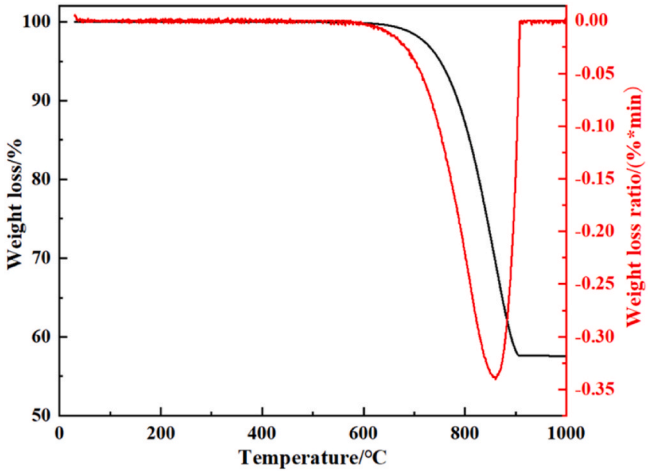


Fig. 9. Plots of the thermogravimetric curve.

Fig. 10(a), (b), and (c) display the original rock sample areas characterized by a relatively uniform distribution of components and structures, featuring small grains. No apparent fractures are visible on their surfaces, although micro-scale pores are present. Fig. 10(d), (e), and (f) display microscopic images of the heat-affected zone where the rock undergoes chemical changes due to high temperature. Specifically, $CaCO_3$ decomposes to form CaO under these elevated temperatures. The entire area appears relatively rough, with noticeable cracks and breakage, along with rock fragments of various sizes. Furthermore, the elevated temperature and thermal stress have caused some of the rock pieces to fragment into smaller particles, which are dispersed in the vicinity of the cracks. Following laser irradiation, the high thermal stress causes the internal components of the rock sample to expand and fracture, resulting in the appearance of microfractures. As heat continues to accumulate, these fractures progressively widen and form noticeable cracks.

3.6. Variation of pore distribution

Laser irradiation significantly affects the T_2 distribution curve in the saturated water state, as depicted in Fig. 11. The T_2 relaxation time distribution curve is typically classified as triple-peak, double-peak, or single-peak, based on the number of wave peaks. The test result for this specimen indicates a double-peak curve. To distinguish the wave peaks of the T_2 distribution curves before and after laser irradiation, this paper uses P_1^b and P_2^b to represent the two wave peaks before microwave, P_1^a and P_2^a to represent the two wave peaks after microwave treatment. It's worth noting that the distinction between crests and peaks typically depends on the presence of a distinct trough between the crest and the wave. If no such trough exists, the two peaks are considered as one wave. The magnitude of the relaxation amplitude within the trough between the wave crest and wave peak can serve as an indicator of the connectivity between different pore types. The relaxation time range in the T_2 distribution curve changes after laser irradiation, with larger relaxation times corresponding to larger pore sizes. It is clear from Fig. 14 that the wave peak of P_1^a is larger than that of P_1^b and the lower limit and span of the relaxation time increase after laser irradiation, and the wave peak of P_2^a is larger than that of P_2^b and the lower limit and span of the relaxation time increase. It means that the radius of some of the pore structures in the specimen increases and the number of pores increases. Following laser irradiation, the specimens initially exhibit minor internal damage, which accumulates progressively with continued irradiation. Consequently, the damage to the pore structure becomes increasingly severe over time. The irradiation of the specimen leads to changes in the pore structure, the formation of a large number of cracks, and even the rupture into fragments. Hence, laser irradiation leads to an enlargement of the pore radius within the specimen, signifying that it can enhance the gas transport channels within the rock, thereby aiding in oil and gas recovery efforts.

In Fig. 12(a), the distribution of small hole diameters in the specimen before laser irradiation is depicted. It is evident that the two intervals, 0.0025–0.01 μm and 0.01–0.025 μm, represent the primary size ranges for small holes, comprising 37.88 % and 40.64 % of the total, respectively, making up nearly 80 % in total. Fig. 12(b) shows the distribution of the small hole diameter of the specimen after laser irradiation. Indeed, it's evident that small pores occupy 18.85 % and 18.08 % within the size range of 0.0025–0.01 μm and 0.01–0.025 μm, respectively, amounting to a total percentage slightly exceeding 35 %. After laser irradiation, the proportion of holes in the large and medium holes is uniformly distributed, and the total proportion is more than 50 %. When

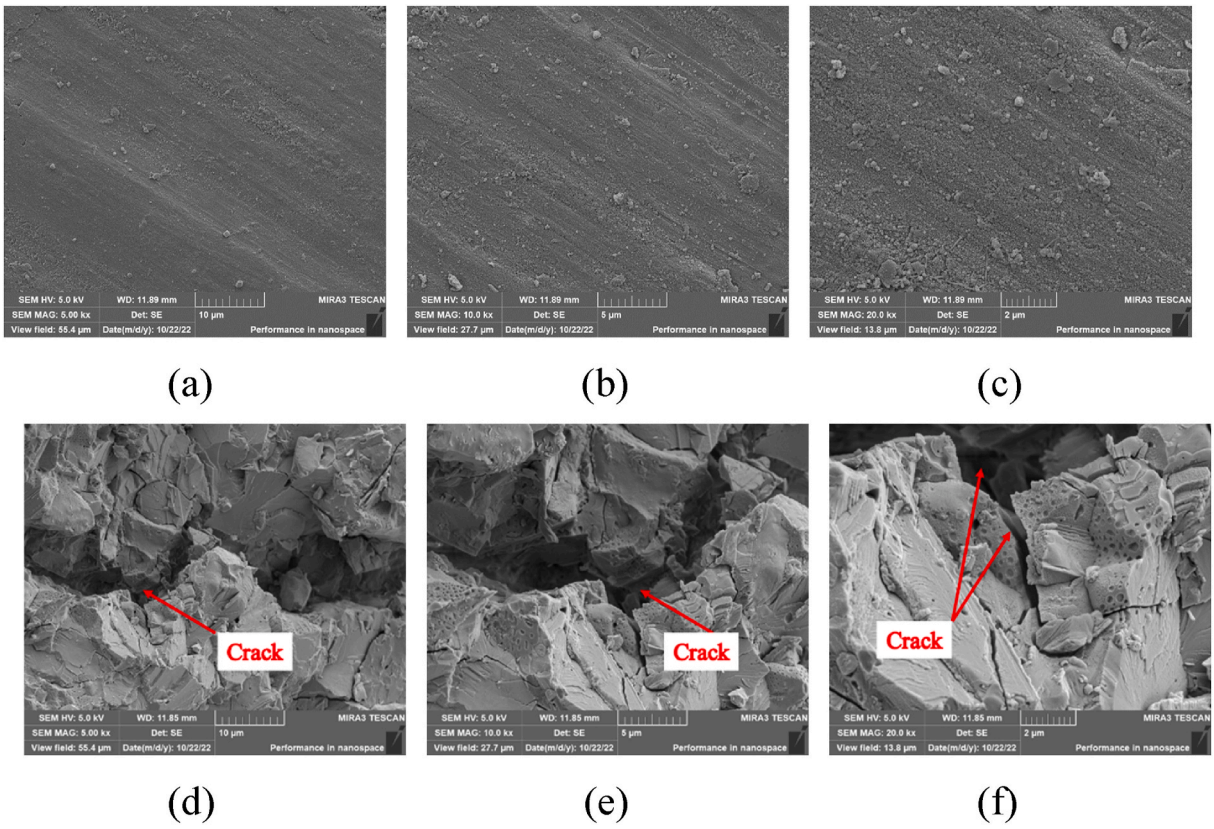


Fig. 10. Images for SEM (a) laser non-irradiation area (× 5 kx); (b) laser non-irradiation area (× 10 kx); (c) laser non-irradiation area (× 20 kx); (d) laser irradiation area (× 5 kx); (e) laser irradiation area (× 10 kx); (f) laser irradiation area (× 20 kx).

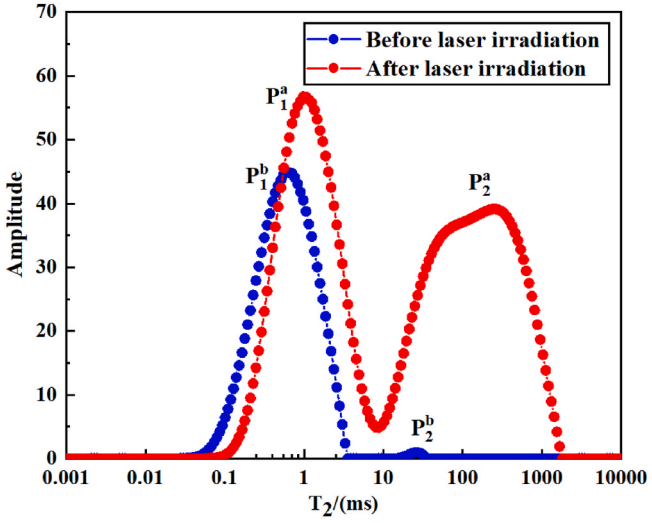


Fig. 11. Effect of laser irradiation on NMR T_2 distribution in saturated water.

comparing the distribution of pore diameters before and after laser irradiation, it is evident that the pore diameter of the rock increases after laser irradiation and becomes more widely distributed. This is advantageous for facilitating fracture expansion and the transport of oil and gas. When comparing Fig. 12(a) and (b), it becomes evident that the diameter of small holes in the rock increases after laser irradiation, and their distribution becomes more widespread. This change is more conducive to fracture expansion and the transport of oil and gas.

3.7. Load–displacement curves

Fig. 13 is a schematic diagram of nanoindentation, illustrating the specific positioning of the nanoindentation. Throughout the nanoindentation test, a sensor continuously monitors the indenter load and the depth of indenter penetration into the rock surface, thereby generating a load-displacement curve. This curve offers insights into the localized deformation progression of the sample surface under the applied load. Indeed, nanoindentation testing furnishes valuable data on parameters such as contact depth, response to applied force, and creep deformation exhibited by both the sample and the indenter tip during the loading and unloading phases. These data serve as the basis for calculating the elastic modulus and indentation hardness values of the tested sample. Also, the load-displacement curve reflects the microstructural information of the sample.

Fig. 14 depicts the load-displacement curves from the nanoindentation test on the rock section. These curves consist of three segments: the loading stage, holding stage, and unloading stage. The varying slopes in the loading stage between the curves indicate distinctions in the mechanical response of different regions of the rock surface when subjected to indenter loading. This variation can be attributed to the somewhat random distribution of indentation points on the rock surface, and the mechanical response of each indentation point during the nanoindentation test varies because of the highly heterogeneous nature of the rock. During the holding stage, the horizontal span of the curve signifies the extent of ongoing indenter penetration at maximum load, effectively reflecting the creep characteristics of the rock surface at various indentation points under the indenter load. Fig. 14(a) illustrates the load-displacement diagram for the rock without laser irradiation. The deformation of the rock surface exhibits significant variations at different indentation points, with the maximum penetration depth of the indenter ranging from 180 to 630 nm. Fig. 14(b) shows

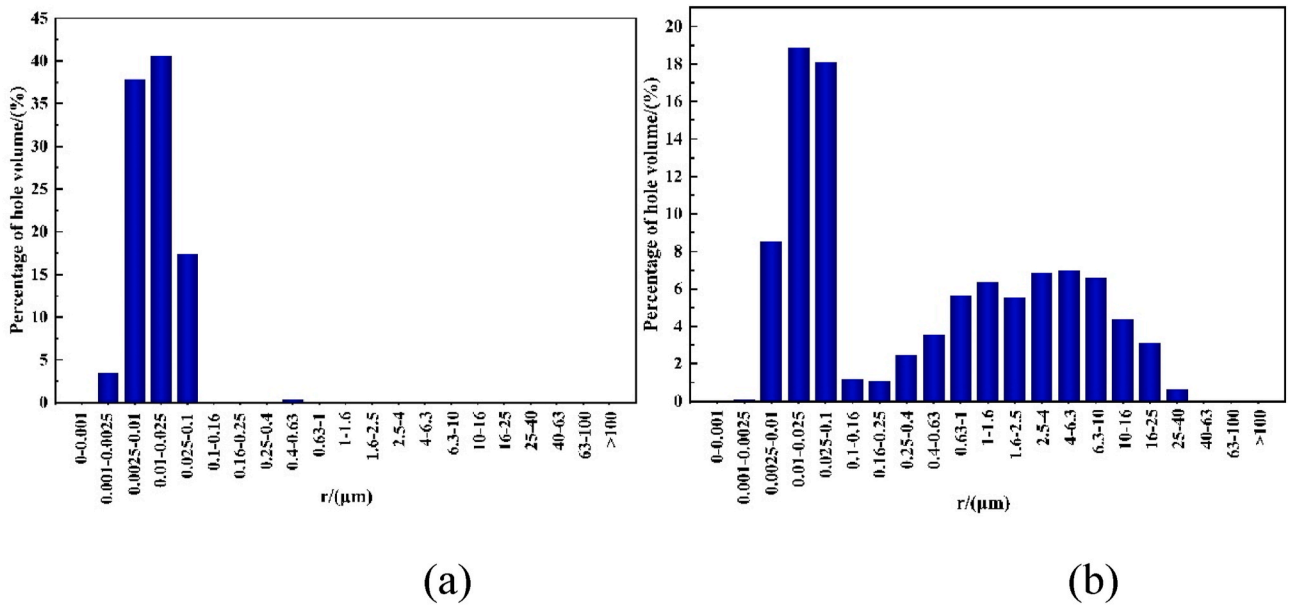


Fig. 12. Histogram of small hole distribution (a) before laser irradiation; (b) after laser irradiation.

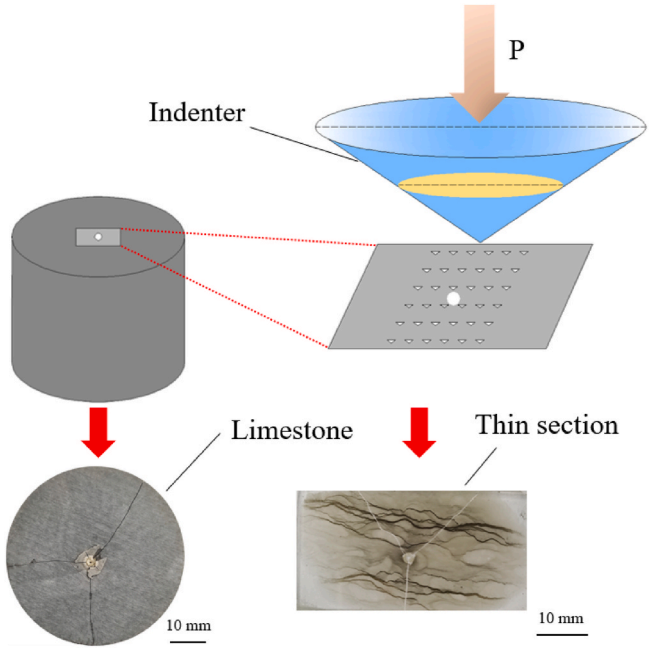


Fig. 13. Nanoindentation schematic diagrams.

the load-displacement diagram of the laser irradiation area. Indeed, the deformation at various indentation points on the rock surface exhibits significant variation, and the maximum penetration depth of the indenter ranges from 180 to 1200 nm. Differences in the range of maximum penetration depths are related to the mineralogical diversity of the rocks. Chemical reactions take place in rocks irradiated by lasers, leading to the decomposition of the native mineral calcite into carbon dioxide and calcium oxide, resulting in differences from the original mineral. The laser-irradiated rock undergoes a chemical reaction in which calcium carbonate compounds break down to form carbon dioxide and calcium oxide, resulting in differences from the native minerals. Furthermore, the load-displacement curve characterized by a smaller penetration depth during the loading stage (i.e., the curve with a steeper slope during loading) corresponds to less deformation during the holding stage and also exhibits the smallest maximum penetration depth. The smoothest load-displacement curve observed during the loading stage tends to exhibit greater creep and a larger maximum penetration depth during the holding stage. The mechanical strength of the rock surface minerals corresponding to these two load-displacement curves should differ significantly.

During the loading and holding stages, the indenter continues to penetrate the rock surface, which leads to plastic deformation of the rock surface. To study the effect of laser irradiation on the elastic-plastic deformation during nanoindentation testing, the variation of the maximum indentation depth h_{max} of rock samples during indentation testing is compared. During the loading stage, the applied pressure induces elastic-plastic deformation at the local surface of the rock, with the indentation depth reaching its maximum before the initial unloading. The range of maximum indentation depth variation in the laser non-irradiation area and the laser irradiation area is shown in Fig. 15. The maximum indentation depth in the laser non-irradiation area by the laser ranges from 171.18 to 666.38 nm, and the average value of the maximum indentation depth is 295.83 nm. Following laser irradiation, the maximum indentation depth in this region ranges from 194.75 to 1916.08 nm, with an average maximum indentation depth of 708.23 nm. It is evident that there exists a substantial disparity in the maximum indentation depth levels between the laser non-irradiated area and the laser irradiated area under the same load. The region subjected to laser irradiation exhibits noticeably greater deformation during the nanoindentation test, featuring a 139.4 % increase in the average maximum indentation depth when compared to the non-laser-irradiated areas. This suggests that the rock's resistance to deformation is diminishing after laser irradiation.

3.8. Elastic modulus and hardness

Nanoindentation tests are conducted in both the laser non-irradiated area and the laser irradiated area. The resulting elastic modulus (E) and surface hardness (H) of the rock are calculated and statistically distributed, as illustrated in Fig. 16. The frames with distinct backgrounds in the figure depict data for the non-irradiated and post-laser irradiated regions, respectively. Within each frame, the solid diamond represents the data points obtained from the nanoindentation test, and the box plot on the right depicts the distribution of the data points. The short lines positioned at the top and bottom of the box denote the upper and lower

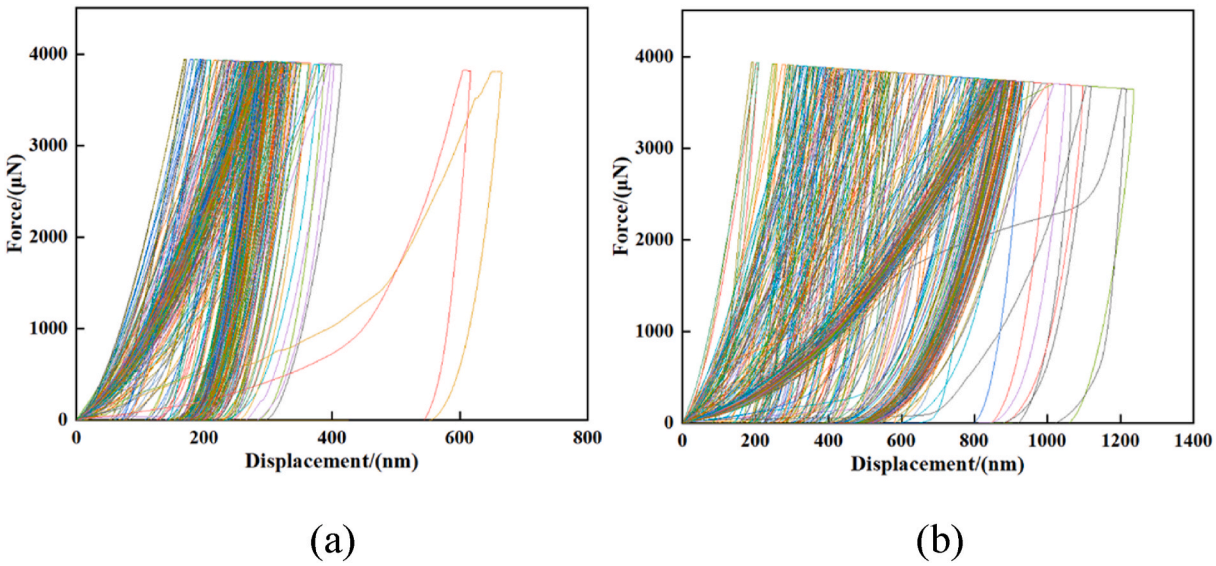


Fig. 14. Load-displacement curves of individual indentation matrices on the rock surface (a) laser non-irradiation area; (b) laser irradiation area.

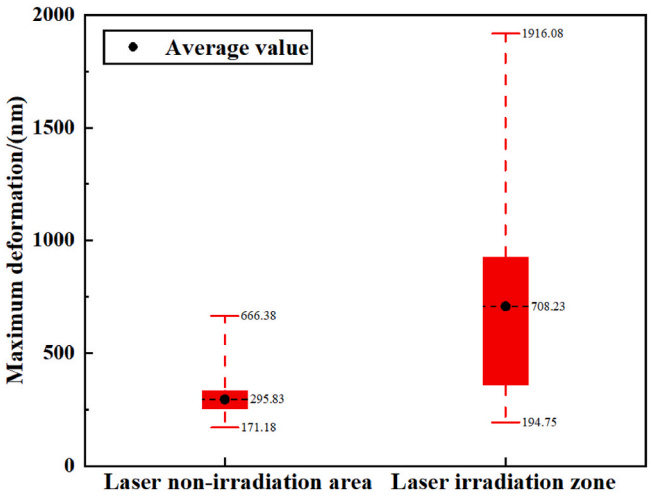


Fig. 15. Maximum indentation depth variation on the rock surface.

limits of the box plot, respectively. Within the boxes, the black dots within the circles represent the average values.

Fig. 16(a) shows the variation of microscopic elastic modulus in the laser non-irradiation area and the laser irradiation area. The interval of the microscopic elastic modulus of the laser non-irradiation area ranges from 16.21 to 72.34 GPa, and the average value of the microscopic elastic modulus is 36.69 GPa. The range of microscopic elastic modulus values for rocks after laser irradiation spans from 3.22 to 52.03 GPa, with an average value of 10.06 GPa. The average microscopic modulus of elasticity of rocks irradiated with laser is reduced by 72.58 % compared to that of laser non-irradiation area. Fig. 16(b) shows the changes in the hardness in the laser non-irradiation area and the laser irradiation area. The hardness range of rocks without laser irradiation falls between 0.23 and 8.44 GPa, with an average hardness of 1.67 GPa. The range of rock hardness after laser irradiation spans from 0.03 to 5.17 GPa, with an average hardness value of 0.47 GPa. The average hardness of the rocks is reduced by 71.86 % compared to the average hardness of the laser non-irradiation area. The microscopic modulus of elasticity and hardness of the rocks after laser irradiation exhibit a substantial decrease, signifying that laser irradiation has a profoundly destructive impact on the rock samples. Furthermore, these discrete data values reflect the significantly non-homogeneous nature of the microscopic properties of the rock samples.

Fig. 17 compares the variations in the mechanical properties of rocks tested through nanoindentation, all on the same coordinate axis. The elastic modulus of laser non-irradiation area is concentrated in the range of hardness 0–3 GPa, presenting a wide shape, as shown in Fig. 17(a). The microscopic elastic modulus of rocks exposed to laser irradiation is predominantly clustered within the 0–2 GPa range, displaying a narrow distribution, as illustrated in Fig. 17(b). This phenomenon occurs due to the reduction in the mechanical properties of the rock following laser irradiation, resulting in a pronounced concentration of these properties.

4. Discussion

High-density laser beams generate rapid temperature increases on the rock surface. The elevated temperature leads to alterations in the mechanical and physical properties of the rock. There are three forms of rock damage caused by laser irradiation: thermal fracture, melting, and evaporation,[30,33,37] as shown in Fig. 18.

Thermal fracture: During heat conduction, temperature variations arise within the mineral grains within the rock. The expansion of a mineral particle that absorbs more heat leads to the compression of adjacent mineral particles that absorb less heat, consequently generating stress. Due to this asynchronous deformation among mineral grains, thermal stresses are generated. Cracks are induced when these stresses surpass the tensile strength of the rock's mineral grains. This process may also magnify pre-existing microfractures in the rock. During this phase, thermal fractures initiate and propagate from the irradiation center.

SEM results (Fig. 10) reveal that the rocks within the laser irradiation zone exhibit more pronounced fractures and more extensive pore development when compared to those in the original rock zone. Pore distribution tests are conducted on intact rock samples both before and after laser irradiation, utilizing nuclear magnetic resonance equipment (Fig. 11). The results indicate a substantial increase in the relaxation time T_2 and a significant elevation in wave height for the specimens subjected to laser irradiation. Furthermore, there is a marked increase in the distribution of the proportion of larger and medium-sized pores, accounting for more than 50 % of the total pore proportion.

Melting: Once the mineral reaches its melting point, the mineral particles transition into the liquid phase. During laser irradiation of rocks, calcium carbonate compounds (such as calcite and dolomite)

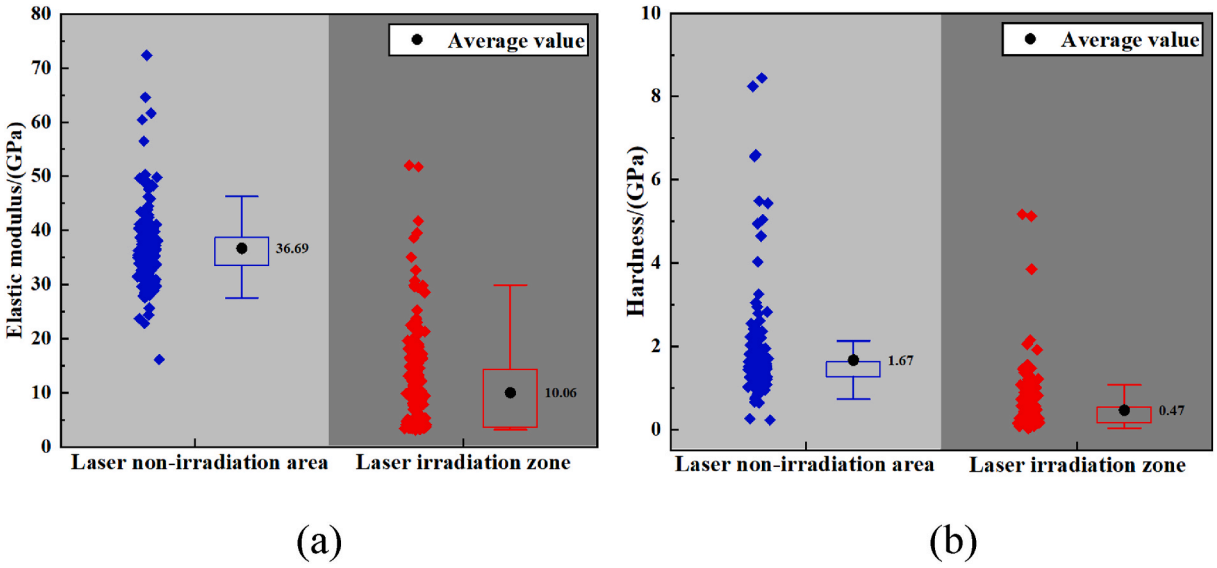


Fig. 16. (a) Distribution of elastic modulus in a different zone; (b) distribution of hardness in a different zone.

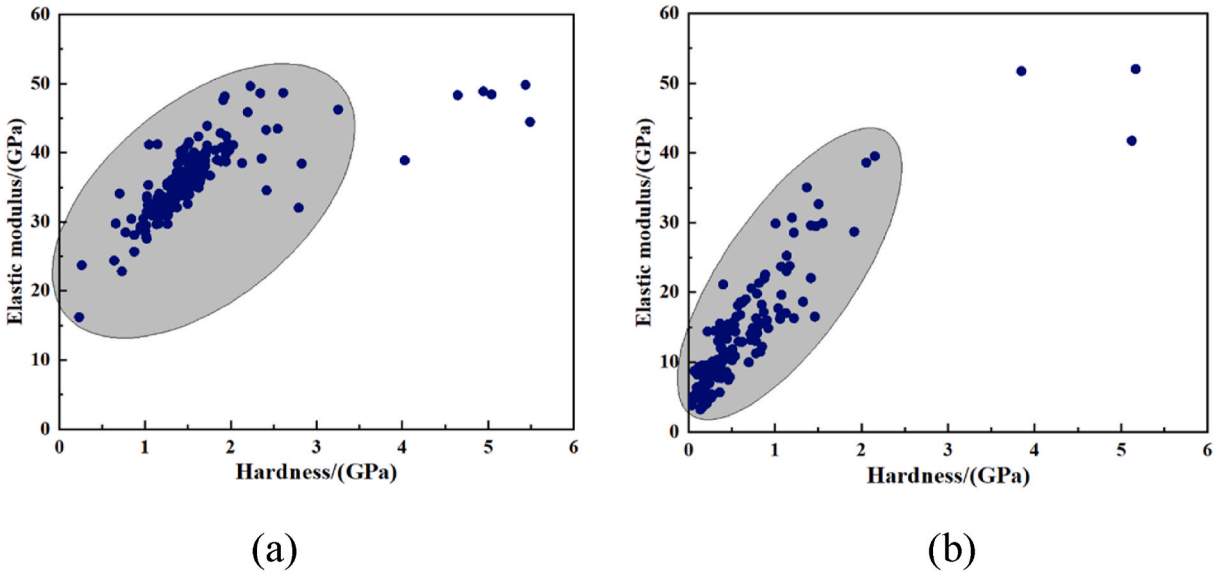


Fig. 17. Evolution of microscopic mechanical properties of rock specimens (a) laser non-irradiation area; (b) laser irradiation zone.

undergo thermal decomposition at elevated temperatures, producing carbon dioxide and calcium oxide. In the laser irradiation zone, both calcite and dolomite within the rocks are reduced, leading to the formation of $Ca(OH)_2$ and CaO. Furthermore, the presence of C and O elements in the rocks of the irradiated areas diminishes due to the release of CO_2.

Assuming that a laser beam of power P is uniformly irradiated on the rock and melting occurs, then the removal rate u of rock melting is as follows[46]:

$$u=\frac{\eta P-\oint_{S_1}K|\nabla T|ds+\oint_{S_2}\varepsilon\sigma\left(T_m^4-T_0^4\right)ds}{\pi a^2(M+CT_m)\rho} \tag{4}$$

where η refers to the absorption rate, K is the heat transfer coefficient, σ is the Stepan constant, T refers to the absolute temperature of the rock, T_m refers to the melting point of the rock, T_0 refers to the initial temperature of the rock, M refers to the latent heat of melting, and C refers to the specific heat capacity of the rock. $(M+CT_m)\rho/\eta$ is the specific energy value, representing the amount of heat required to melt a unit volume of rock.

Evaporation: Due to the high laser power and prolonged irradiation, the temperature of the rock reaches a critical point, leading to evaporation. When a laser beam with power P uniformly irradiates the rock, and evaporation occurs, the removal rate (u) of rock evaporation can be expressed as follows[47]:

$$u=\frac{\eta P-\oint_{S_1}K|\nabla T|ds+\oint_{S_2}\varepsilon\sigma\left(T_g^4-T_0^4\right)ds}{\pi a^2\left(G+M+CT_g\right)\rho} \tag{5}$$

where G represents the latent heat of evaporation of the rock; and T_g represents the critical temperature at which evaporation of the rock occurs.

5. Conclusions

In this study, the composition of compounds and elements in the laser-irradiated rocks is analyzed using XRD and EDS. Additionally, the

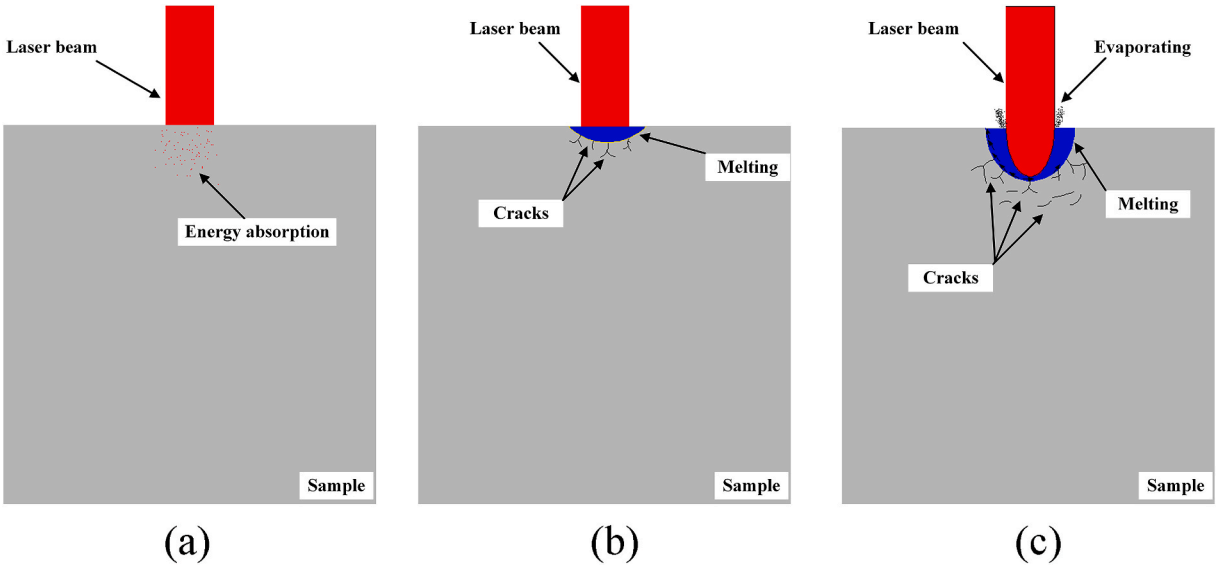


Fig. 18. Schematic diagram of rock breaking by laser irradiation. (a) Thermal fracture; (b) Melting; (c) Evaporation.

rocks' weight loss temperature is determined through TGA. The changes of rock microstructure, pore number and pore structure after laser irradiation are observed by SEM and NMR. Quantitative analysis of changes in the micromechanical properties, such as elastic modulus and hardness, of rocks after laser irradiation is performed using the nano-indentation technique. According to the experimental results, the conclusions are as follows.

(1) The primary mineral in both the original rock sample and the sample irradiated by laser is calcite, with diffraction peaks at 3250 and 1300, respectively. The mineral composition of the laser-irradiated area includes quartz (1.7 %), calcite (26.2 %), portlandite (10.4 %), calcium oxide (1.8 %), and amorphous minerals (59.9 %). New components such as $Ca(OH)_2$ and CaO are produced in the irradiated area.
(2) The mass percentage of carbon (C) decreases from 10.10 % before irradiation to 9.70 % after irradiation, while the mass percentage of oxygen (O) decreases from 46.38 % before irradiation to 43.99 % after irradiation. Meanwhile, the main weight loss temperature of the sample occurs between 640 °C and 900 °C. The decomposition minerals in the rock samples in this temperature range are mainly calcite and dolomite.
(3) The internal composition of the rock sample appears swollen and exhibits microfractures when examined under a scanning electron microscope (SEM) after laser irradiation. The percentage of large and medium holes is uniformly distributed with a total percentage of more than 50 % after laser irradiation as analyzed by NMR. The small holes occupy 18.85 % and 18.08 % in the interval of 0.0025–0.01 μm and 0.01–0.025 μm, respectively, with a total percentage of just over 35 %.
(4) The maximum indentation depth in the laser non-irradiation area by the laser ranges from 171.18 to 66638 nm, and the average value of the maximum indentation depth is 295.83 nm. The maximum indentation depth of the area after laser irradiation ranges from 194.75 to 1916.08 nm, and the average value of the maximum indentation depth is 708.23 nm.
(5) The average microscopic modulus of elasticity of the rocks decreased significantly from 36.69 GPa without laser irradiation to 10.06 GPa with laser irradiation, representing a reduction of approximately 72.58 %. The average hardness of the rock is reduced from 1.67 GPa without laser irradiation to 0.47 GPa with laser irradiation, which is a decrease of 71.86 %.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled.

Data availability

Data will be made available on request.

Acknowledgments

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